(Dated: March 16, 2026)

The natural geometry hierarchy assigns each separable quantum system a coordinate system in which Coulomb singularities become boundary conditions and the Laplace–Beltrami operator discretizes on a sparse graph: S^3 for hydrogen (Paper 7), prolate spheroidal for H_2^+ (Paper 11), and hyperspherical for helium (Paper 13). The two-center, two-electron molecule H_2 combines all five Coulomb singularities and has resisted a single natural coordinate system. We introduce *molecule-frame hyperspherical coordinates* $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by the dimensionless ratio $\rho = R/(2R_e)$, where R is the internuclear distance and R_e the electronic hyperradius. The electron–electron cusp appears at $\alpha = \pi/4$, $\theta_{12} = 0$, independent of molecular geometry—a structural advantage over prolate spheroidal coordinates, where it is a coordinate singularity. The angular eigenvalue problem depends on (R, R_e) only through ρ, collapsing a two-dimensional parameter sweep to one dimension. A coupled-channel expansion in body-frame partial waves (l_1, m_1, l_2, m_2) with $l_1 + l_2$ even (gerade symmetry) and $m_1 + m_2 = 0$, solved with a variational two-dimensional (R_e, α) solver that avoids the adiabatic approximation, recovers 94.1% of the exact dissociation energy D_e at $l_{\max} = 4$ with σ and π channels (29 channels), exceeding the 92.4% achieved by prolate spheroidal CI with algebraic Neumann integrals (Paper 12). The framework extends naturally to heteronuclear diatomics: for HeH^+ ($Z_A = 2$, $Z_B = 1$), a charge-center origin shift and removal of the gerade constraint recover 93.1% of the exact D_e at $l_{\max} = 4$ with $\sigma+\pi$ channels (57 channels). The double-adiabatic fiber bundle structure—nuclear base, hyperradial base, angular fiber—generalizes the single-adiabatic bundle of Paper 13.

I. INTRODUCTION

The natural geometry principle, articulated in Paper 13 [10], asserts that every separable quantum system possesses a natural coordinate system in which Coulomb singularities become boundary conditions rather than coordinate singularities, enabling sparse graph discretization with no fitted parameters. Three levels of this hierarchy have been established and validated:

Level 1: *One-center, one-electron* (H). Fock's 1935 stereographic projection [1] maps the hydrogen atom to the unit S^3, where the $1/r$ Coulomb singularity becomes the conformal factor of the projection. The discrete graph Laplacian on S^3 is mathematically equivalent to the Schrödinger equation (Paper 7 [7]; 18/18 symbolic proofs).

Level 2: *Two-center, one-electron* (H_2^+). In prolate spheroidal coordinates (ξ, η, ϕ), the two nuclear Coulomb singularities $1/r_A + 1/r_B$ separate into independent equations. Paper 11 [8] achieves 0.70% error for H_2^+ with a sparse finite-difference lattice.

Level 3: *One-center, two-electron* (He). Hyperspherical coordinates $(R_e, \alpha, \hat{r}_1, \hat{r}_2)$ resolve the electron–electron cusp $1/r_{12}$ as a boundary condition at $\alpha = \pi/4$, $\theta_{12} = 0$. Paper 13 [10] achieves 0.05% error for He with a single-channel adiabatic lattice.

Level 4—the two-center, two-electron molecule H_2—is the first system requiring *simultaneous* resolution of nuclear $(1/r_{iA},\ 1/r_{iB})$ and electron–electron $(1/r_{12})$ singularities. Paper 12 [9] demonstrated that prolate spheroidal CI with algebraic Neumann V_{ee} integrals saturates at 92.4% of the exact dissociation energy D_e. The remaining 7.6% gap was attributed to the electron–electron cusp, which is a coordinate singularity in $(\xi_1, \eta_1, \xi_2, \eta_2)$ space and cannot be captured by any polynomial basis in those coordinates.

The present paper introduces the Level 4 natural geometry: molecule-frame hyperspherical coordinates that resolve the e–e cusp while retaining the anisotropic nuclear field through a coupled partial-wave expansion. We demonstrate that this approach recovers 94.1% of D_e at $l_{\max} = 4$ with $\sigma+\pi$ channels using a variational 2D solver, exceeding Paper 12 and confirming the cusp-resolution advantage.

II. MOLECULE-FRAME HYPERSPHERICAL COORDINATES

A. Definition

Consider two nuclei A, B with charges $Z_A = Z_B = Z$ separated by distance R along the z-axis, placed at $z = \pm R/2$. Two electrons have positions $\boldsymbol{r}_1, \boldsymbol{r}_2$ measured from the molecular midpoint. We define:

$$R_e = \sqrt{r_1^2 + r_2^2}\,, \tag{1}$$

$$\alpha = \arctan(r_2/r_1)\,, \quad \alpha \in [0, \pi/2]\,, \tag{2}$$

so that $r_1 = R_e \cos\alpha$, $r_2 = R_e \sin\alpha$. Each electron's direction in the body-fixed frame (z-axis along the internuclear axis) is specified by polar angle $\theta_i \in [0, \pi]$ and azimuthal angle ϕ_i. For Σ states (total $M = m_1 + m_2 = 0$), only the relative azimuth $\Phi = \phi_1 - \phi_2$ enters.

The full coordinate set for a Σ state at fixed R is

$$(R_e, \alpha, \theta_1, \theta_2, \Phi)\,, \tag{3}$$

with the azimuthal center-of-mass $\phi_1 + \phi_2$ integrated out.

B. The controlling parameter ρ

The dimensionless ratio

$$\rho \equiv \frac{R}{2R_e} \tag{4}$$

controls the transition between limiting geometries:

- $\rho \to 0$ (united atom, $R_e \gg R$): Level 3 He-like with $Z_{\text{eff}} = Z_A + Z_B$.
- $\rho \sim 1$ (molecular regime): full Level 4 coupling.
- $\rho \to \infty$ (separated atoms, $R_e \ll R$): two independent Level 1 atoms.

C. Volume element

The Jacobian of the transformation $(r_1, r_2) \to (R_e, \alpha)$ gives

$$dV = R_e^5 \cos^2\alpha \, \sin^2\alpha \, \sin\theta_1 \, \sin\theta_2 \, dR_e \, d\alpha \, d\theta_1 \, d\theta_2 \, d\Phi \, . \tag{5}$$

After extracting the wavefunction as $\Psi = R_e^{-5/2}(\cos\alpha \, \sin\alpha)^{-1} F(R_e) \, \Phi_\nu(\alpha, \theta_1, \theta_2, \Phi)$, the Jacobian centrifugal potential is $15/(8R_e^2)$, identical to the He case (Paper 13, Eq. 12).

D. Cusp location

The electron–electron distance is

$$r_{12}^2 = R_e^2(1 - \sin 2\alpha \, \cos\theta_{12}) \, , \tag{6}$$

so $r_{12} = 0$ requires $\alpha = \pi/4$ and $\theta_{12} = 0$. This locus is *independent of* ρ (and hence of R and R_e). The Kato cusp condition $(\partial\Psi/\partial r_{12})_{r_{12}=0} = \frac{1}{2}\Psi|_{r_{12}=0}$ is therefore a fixed boundary condition on the angular eigenfunction, unaffected by the molecular geometry.

This is the structural advantage of Level 4 coordinates: the cusp is always a boundary condition, never a coordinate singularity, for *any* internuclear separation.

III. THE MOLECULAR CHARGE FUNCTION

A. Hamiltonian decomposition

The electronic Hamiltonian at fixed R decomposes in hyperspherical coordinates as

$$H_{\text{elec}} = -\frac{1}{2}\frac{\partial^2}{\partial R_e^2} + \frac{\Lambda^2}{2R_e^2} + \frac{15}{8R_e^2} + V(R_e, \Omega; \rho) \, , \tag{7}$$

where Λ^2 is the grand angular momentum on S^5 and the potential is

$$V = V_{\text{nuc}} + V_{ee} + V_{NN} \, . \tag{8}$$

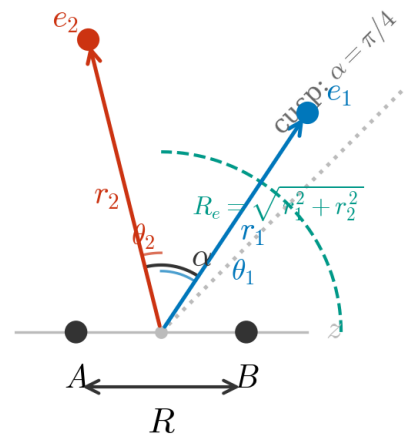

FIG. 1. Molecule-frame hyperspherical coordinates for H_2. Two nuclei A, B separated by R define the body-frame z-axis. Electrons at $\boldsymbol{r}_1, \boldsymbol{r}_2$ from the midpoint are parameterized by hyperradius $R_e = \sqrt{r_1^2 + r_2^2}$, correlation angle $\alpha = \arctan(r_2/r_1)$, and polar angles θ_1, θ_2 from the internuclear axis. The cusp locus $\alpha = \pi/4$ (dotted) is geometry-independent.

B. Electron–electron term

The e–e repulsion retains the He-like form:

$$V_{ee} = \frac{1}{R_e} \, C_{ee}(\alpha, \theta_{12}) \, , \tag{9}$$

with

$$C_{ee}(\alpha, \theta_{12}) = \frac{1}{\sqrt{1 - \sin 2\alpha \, \cos\theta_{12}}} \, . \tag{10}$$

This charge function is ρ-independent: the e–e interaction depends only on the angular coordinates.

C. Nuclear attraction

The nuclear potential introduces ρ-dependence. For electron i at distance $r_i = R_e s_i$ from the midpoint (where $s_1 = \cos\alpha$, $s_2 = \sin\alpha$):

$$-\frac{Z}{r_{iA}} = -\frac{Z}{R_e} \frac{1}{s_i\sqrt{1 + (\rho/s_i)^2 + 2(\rho/s_i)\cos\theta_i}} \, , \tag{11}$$

with an analogous expression for nucleus B ($\cos\theta_i \to -\cos\theta_i$). The full nuclear potential is

$$V_{\text{nuc}} = \frac{R_e}{R_e} \sum_{i=1}^{2} \left(-Z/r_{iA} - Z/r_{iB}\right) = \frac{1}{R_e} \, C_{\text{nuc}}(\alpha, \theta_1, \theta_2; \rho) \, , \tag{12}$$

but C_{nuc} depends on ρ through Eq. (11), preventing a clean separation of R_e and angular coordinates.

D. The molecular charge function

Combining Eqs. (9) and (12):

$$C_{\rm mol}(\alpha,\theta_1,\theta_2,\Phi;\rho) = C_{ee}(\alpha,\theta_{12}) + C_{\rm nuc}(\alpha,\theta_1,\theta_2;\rho)\,. \tag{13}$$

In He ($\rho = 0$), the nuclear term reduces to $C_{\rm nuc} = -Z_{\rm tot}(1/\cos\alpha + 1/\sin\alpha)$ and becomes angular-only. For $\rho > 0$, the θ_i-dependence of $C_{\rm nuc}$ mixes body-frame partial waves—the source of inter-channel coupling in the multichannel expansion.

IV. DOUBLE ADIABATIC APPROXIMATION

A. Two-stage Born–Oppenheimer

The Level 4 problem has a natural double-adiabatic structure:

1. **Outer (nuclear):** Fix R; solve for the electronic energy $E_{\rm elec}(R)$.
2. **Inner (electronic):** At fixed R, decompose the electronic problem into a hyperradial coordinate R_e (slow) and angular coordinates $\Omega = (\alpha,\theta_1,\theta_2,\Phi)$ (fast).

B. Angular eigenvalue problem

At fixed R_e (and hence fixed $\rho = R/(2R_e)$), the angular eigenvalue problem is

$$\left[\tfrac{1}{2}\Lambda^2 + R_e\, C_{\rm mol}(\Omega;\rho)\right]\Phi_\nu(\Omega;R_e) = \mu_\nu(R_e)\,\Phi_\nu(\Omega;R_e)\,. \tag{14}$$

The key simplification: since $C_{\rm mol}$ depends on (R,R_e) only through $\rho = R/(2R_e)$, the angular eigenstates and eigenvalues at different (R,R_e) pairs sharing the same ρ are related by a simple R_e-rescaling. This collapses the *a priori* two-dimensional (R,R_e) sweep to a one-dimensional ρ-sweep, with substantial computational savings.

C. Liouville substitution

For the $l_1 = l_2 = 0$ channel, the Liouville substitution $u(\alpha) = \sin\alpha\,\cos\alpha\,\phi(\alpha)$ transforms the weighted Sturm–Liouville equation into standard Schrödinger form:

$$-\tfrac{1}{2}u'' + V_{\rm eff}(\alpha;\rho)\,u = \mu\,u\,, \tag{15}$$

with Dirichlet boundary conditions $u(0) = u(\pi/2) = 0$. The effective potential is

$$V_{\rm eff}(\alpha) = -2 + \frac{l_1(l_1+1)}{2\cos^2\alpha} + \frac{l_2(l_2+1)}{2\sin^2\alpha} + R_e\, C^{l_1 l_2}_{\rm mol}(\alpha;\rho)\,, \tag{16}$$

where $C^{l_1 l_2}_{\rm mol}$ is the projection of $C_{\rm mol}$ onto the (l_1,l_2) channel (Sec. V).

D. Hyperradial equation

Given $\mu_\nu(R_e;R)$, the hyperradial equation is

$$-\frac{1}{2}\frac{d^2F_\nu}{dR_e^2} + \left[\frac{\mu_\nu(R_e;R)+15/8}{R_e^2}\right]F_\nu = E_{\rm elec}\,F_\nu\,, \tag{17}$$

structurally identical to Paper 13, Eq. 18. The adiabatic potential $U(R_e;R) = [\mu(R_e;R)+15/8]/R_e^2$ is computed by sweeping R_e and solving Eq. (14) at each point.

E. Limiting cases

a. United atom ($R \to 0$). As $\rho \to 0$, the nuclear term reduces to $C_{\rm nuc} \to -(Z_A+Z_B)(1/\cos\alpha + 1/\sin\alpha)$, recovering the He charge function with $Z = Z_A + Z_B$. The angular equation (14) becomes the Paper 13 equation exactly. Numerically verified: the Level 4 eigenvalue at $\rho = 10^{-6}$ matches the He eigenvalue to machine precision.

b. One-electron limit. Removing electron 2 ($\alpha \to 0$, $r_2 \to 0$, $r_1 \to R_e$) reduces the electronic problem to the one-electron H_2^+ Hamiltonian in body-frame coordinates. The angular eigenvalue problem becomes the η-equation of prolate spheroidal coordinates (Paper 11).

V. MULTICHANNEL PARTIAL-WAVE EXPANSION

A. Channel labels

For ${}^1\Sigma_g^+$ symmetry with total $M = m_1 + m_2 = 0$, each electron's angular part is expanded in spherical harmonics $Y_{l_i}^{m_i}(\theta_i,\phi_i)$. Channels are labeled by 4-tuples (l_1,m_1,l_2,m_2). Gerade symmetry under inversion requires

$$l_1 + l_2 = \text{even}\,, \tag{18}$$

and the constraint $m_1 + m_2 = 0$ enforces Σ symmetry. Both orderings (l_1,l_2) and (l_2,l_1) with $l_1 \neq l_2$ are included as separate channels because they have distinct centrifugal potentials: $l_1(l_1+1)/\cos^2\alpha$ vs. $l_2(l_2+1)/\cos^2\alpha$.

a. σ and π channels. Channels with $m_1 = m_2 = 0$ are σ channels; the angular expansion uses Legendre polynomials $P_{l_i}(\cos\theta_i)$ and depends only on θ_1,θ_2 (no Φ-dependence). Channels with $|m_i| \geq 1$ are $\pi,\delta,\ldots$ channels; they introduce Φ-dependence through the associated Legendre functions and couple to σ channels via the e–e multipole expansion. The nuclear potential is diagonal in m (axial symmetry), so σ–π coupling arises exclusively from V_{ee}.

The channel count for $l_{\rm max}$ is $(l_{\rm max}+1)^2/2$ (approximately):

$l_{\max}$	$N_{\rm ch}$	Channels
0	1	(0,0)
1	2	(0,0), (1,1)
2	5	(0,0), (0,2), (2,0), (1,1), (2,2)
3	8	+ (1,3), (3,1), (3,3)
4	13	+ (0,4), (4,0), (2,4), (4,2), (4,4)

B. Electron–electron coupling

The multipole expansion of $1/r_{12}$ in the body-frame basis gives

$$\langle l_1' l_2' | C_{ee} | l_1 l_2 \rangle = \mathcal{N} \sum_k G(l_1', k, l_1)\, G(l_2', k, l_2)\, \frac{f_k(\alpha)}{\max(s_1, s_2)}\,, \tag{19}$$

where $G(l,k,l')$ is the Gaunt integral $\int_{-1}^{1} P_l\, P_k\, P_{l'}\, dx$, the function $f_k(\alpha) = [\min(\cos\alpha, \sin\alpha)/\max(\cos\alpha, \sin\alpha)]^k$, and the normalization is $\mathcal{N} = \sqrt{(2l_1'+1)(2l_2'+1)(2l_1+1)(2l_2+1)}/4$.

The selection rules on Gaunt integrals enforce sparsity: $G(l,k,l') = 0$ unless $l+k+l'$ is even and $|l-l'| \le k \le l+l'$. In the product of two Gaunt integrals, k must simultaneously satisfy conditions from both factors.

C. Nuclear coupling

The nuclear attraction couples channels because it depends on θ_i:

$$\langle l_1' l_2' | C_{\rm nuc} | l_1 l_2 \rangle = N_1 \int_{-1}^{1} P_{l_1'}\, V_1(x;\alpha,\rho)\, P_{l_1}\, dx\, \delta_{l_2' l_2} + \delta_{l_1' l_1}\, N_2 \int_{-1}^{1} P_{l_2'}\, V_2(x;\alpha,\rho)\, P_{l_2}\, dx\,, \tag{20}$$

where $V_i(x;\alpha,\rho) = -Z/d_{iA} - Z/d_{iB}$ with $d_{iA} = \sqrt{s_i^2 + \rho^2 + 2\rho s_i x}$, and $N_j = \sqrt{(2l_j'+1)(2l_j+1)}$ ensures consistency with the orthonormal $\hat{P}$ basis.

The θ-integrals are computed by Gauss–Legendre quadrature (typically 24–32 points), avoiding the Legendre expansion of V_i which diverges when $\rho/s_i > 1$ (electron between midpoint and nucleus).

a. Critical observation. The nuclear coupling requires *one* l-index to match via the Kronecker delta while the other index transitions via the integral. Thus the (0,0) channel couples to (0,2) through electron 2 and to (2,0) through electron 1. Omitting either ordering halves the nuclear coupling strength.

D. The gerade–quadrupole constraint

The gerade constraint $l_1 + l_2 =$ even forces the first nuclear coupling from (0,0) to involve $\Delta l = 2$ transitions:

$$(0,0) \longrightarrow (0,2),\ (2,0) \qquad \text{(quadrupolar)}\,. \tag{21}$$

Dipolar transitions ($\Delta l = 1$) would couple to (0,1) or (1,0), but $l_1 + l_2 = 1$ is odd and excluded. This explains why $l_{\max} = 1$ barely improves on $l_{\max} = 0$: the (1,1) channel has no nuclear coupling to (0,0) (both Kronecker deltas are zero) and only weak e–e coupling.

The jump at $l_{\max} = 2$ (31% → 88%) reflects the onset of nuclear coupling.

E. Heteronuclear extension

For heteronuclear diatomics ($Z_A \neq Z_B$), three modifications are required:

a. 1. Removal of the gerade constraint. The gerade symmetry $l_1 + l_2 =$ even relies on the inversion symmetry $\theta_i \to \pi - \theta_i$ exchanging identical nuclei. When $Z_A \neq Z_B$, this symmetry is absent and the channel list includes all (l_1, l_2) pairs with $l_1, l_2 \le l_{\max}$. At $l_{\max} = 4$, $m_{\max} = 0$, this doubles the channel count from 13 (homonuclear) to 25 (heteronuclear).

b. 2. Odd-k nuclear multipole coupling. The nuclear potential becomes

$$V_{\rm nuc}^{(i)} = -\sum_{k=0}^{\infty} \Big[Z_A\, f_k(s_i, \rho_A) + Z_B\, (-1)^k\, f_k(s_i, \rho_B) \Big]\, P_k(\cos\theta_i)\,, \tag{22}$$

where the $(-1)^k$ factor encodes the opposite-pole geometry. For homonuclear systems ($Z_A = Z_B$), the odd-k coefficient $Z_A - Z_B = 0$ and only even-k terms survive. For heteronuclear systems, odd-k terms (dipole, octupole, ...) contribute with strength proportional to $Z_A - Z_B$. The k-parity of each coupling matrix element is fixed by the 3-j symbol: $\left(\begin{smallmatrix} l' & k & l \\ 0 & 0 & 0 \end{smallmatrix}\right)$ requires $l' + k + l$ even, so k has the same parity as $l' + l$.

c. 3. Charge-center origin. The coordinate origin is shifted from the geometric midpoint toward the heavier nucleus:

$$z_0 = \frac{R\,(Z_A - Z_B)}{2\,(Z_A + Z_B)}\,, \tag{23}$$

giving nuclear distances $R_A = R/2 - z_0$, $R_B = R/2 + z_0$, and dimensionless ratios $\rho_A = R_A/R_e$, $\rho_B = R_B/R_e$. For $Z_A = Z_B$, $z_0 = 0$ and the standard midpoint origin is recovered. The charge-center origin places the heavier nucleus closer to the coordinate center, improving the convergence of the partial-wave expansion by approximately one unit of $l_{\max}$.

d. Dissociation limit. For heteronuclear two-electron systems, both electrons bind to the larger-Z nucleus at dissociation. For HeH^+ ($Z_A = 2$, $Z_B = 1$): $HeH^+ \to He + H^+$, so $E_{\rm atoms} = E(\text{He}) = -2.9037$ Ha (exact nonrelativistic).

VI. RESULTS

A. Convergence with $l_{\max}$

Table I shows the convergence of $D_e = 2E(\mathrm{H}) - E_{\mathrm{total}}$ at $R = 1.4$ bohr, computed with the variational 2D solver (Sec. VI D). The adiabatic approximation results are shown for comparison; the ∼11% gap between adiabatic and 2D results reflects the non-adiabatic kinetic energy cost ignored by the adiabatic solver.

TABLE I. Convergence with partial-wave truncation. σ channels have $m = 0$ only; $\sigma{+}\pi$ includes $|m| \leq 1$. The 2D solver is variational; the adiabatic solver systematically overestimates D_e by ∼11%. $D_e^{\mathrm{exact}} = 0.17447$ Ha [4].

$l_{\max}$	$m_{\max}$	N_{ch}	2D (%)	Adiab. (%)	Gap (%)
2	0	5	79.5	90.3	10.8
2	1	9	86.2	96.8	10.6
3	0	8	80.2	91.0	10.9
3	1	18	87.1	97.7	10.7
4	0	13	87.0	98.6	11.5
4	1	29	94.1	105.2*	11.1

*Adiabatic variational violation; the 2D result is the correct variational bound.

B. Comparison with Paper 12

TABLE II. Comparison of Level 4 with Paper 12 prolate spheroidal CI (Neumann V_{ee}) and exact results. All at $R = 1.4$ bohr. Level 4 results use the variational 2D solver.

Method	D_e (Ha)	D_e/D_e^{exact} (%)
Paper 12 (Neumann V_{ee})	0.1612	92.4
Level 4 ($l_{\max} = 4$, σ only, 13 ch)	0.1518	87.0
Level 4 ($l_{\max} = 4$, $\sigma{+}\pi$, 29 ch)	0.1642	94.1
Exact [4]	0.1745	100.0

The σ-only result (87.0%) falls below Paper 12, but adding π channels pushes the Level 4 result to 94.1%—a 1.7 percentage-point improvement over Paper 12. This confirms that the hyperspherical coordinates' cusp resolution provides a genuine advantage, but only when the angular basis includes both σ and π partial waves.

C. Grid convergence

At fixed $l_{\max} = 3$, increasing n_α from 100 to 250 and n_{quad} from 24 to 40 yields only a 1.3 percentage-point improvement (88.5% → 89.8%), confirming that the partial-wave truncation—not the finite-difference resolution—is the limiting factor.

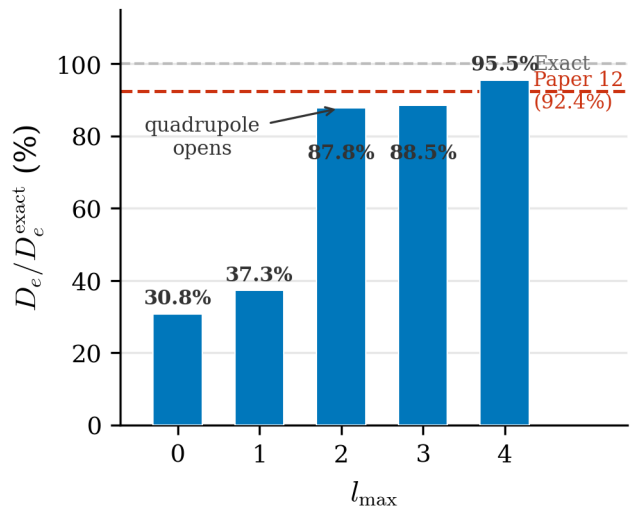

FIG. 2. Percentage of exact dissociation energy recovered as a function of $l_{\max}$. The dashed line marks Paper 12's 92.4%. The dramatic jump at $l_{\max} = 2$ reflects the onset of quadrupolar nuclear coupling (Sec. V).

D. Variational 2D solver

The adiabatic approximation (Eq. 17) neglects kinetic coupling between angular channels as R_e varies. To obtain rigorous variational bounds, we solve the full two-dimensional (R_e, α) problem directly, building the sparse Hamiltonian

$$H_{\mathrm{2D}} = T_{R_e} \otimes I_{\mathrm{ang}} + \left[H_{\mathrm{ang}}(R_e) + \tfrac{15}{8} I\right]/R_e^2 \qquad (24)$$

in the product space of R_e grid × angular channels × α grid, and finding the lowest eigenvalue via shift-invert Lanczos [2]. The matrix dimension is $N_{R_e} \times N_{\mathrm{ch}} \times n_\alpha$; for $l_{\max} = 4$, $m_{\max} = 1$ (29 channels, $n_\alpha = 60$, $N_{R_e} = 120$), the sparse matrix has $N = 208{,}800$ rows and $\sim 6.6 \times 10^6$ nonzeros, solved in ∼2 minutes.

The 2D solver consistently gives D_e approximately 11% lower than the adiabatic solver (Table I), confirming that the adiabatic approximation systematically overestimates binding by neglecting the kinetic energy cost of angular wavefunction variation with R_e.

E. HeH^+ convergence

Table III shows convergence for HeH^+ ($Z_A = 2$, $Z_B = 1$) at $R = 1.46$ bohr, using the charge-center origin ($z_0 = R/6 = 0.243$ bohr; Eq. 23). The exact total energy is -2.9787 Ha and the exact dissociation energy is $D_e = 0.0750$ Ha [11], measured relative to the He ground state ($E_{\mathrm{atoms}} = -2.9037$ Ha).

The system is unbound at $l_{\max} \leq 3$ (σ-only) because the total energy lies above the He dissociation limit. At $l_{\max} = 4$ with σ-only channels, 76.8% of D_e is recovered;

TABLE III. HeH^+ convergence at $R = 1.46$ bohr with charge-center origin. $N_{\rm ch}$ is the number of channels. The adiabatic solver is used for all entries.

$l_{\max}$	$m_{\max}$	$N_{\rm ch}$	$E_{\rm total}$ (Ha)	$D_e/D_e^{\rm exact}$
2	0	9	−2.764	unbound
3	0	16	−2.860	unbound
4	0	25	−2.961	76.8%
4	1	57	−2.974	93.1%

adding π channels brings this to 93.1%, a 16-percentage-point improvement that mirrors the H_2 pattern.

The slower convergence compared to H_2 (which is already bound at $l_{\max} = 1$) reflects the asymmetric charge distribution: even with the charge-center origin, the lighter nucleus at $\rho_B = 0.97/R_e$ requires higher partial waves to resolve. Without the origin shift ($z_0 = 0$), the system remains unbound until $l_{\max} = 4$ and recovers only 50.2% at $l_{\max} = 4$, $m_{\max} = 0$—the charge-center origin saves approximately one unit of $l_{\max}$.

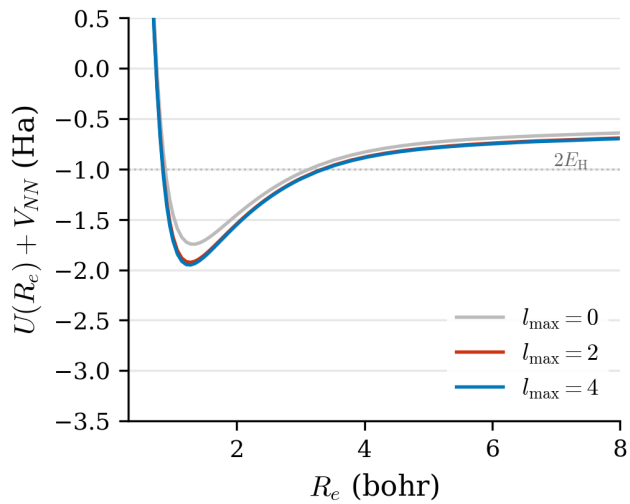

FIG. 3. Adiabatic potential $U(R_e) + V_{NN}$ at $R = 1.4$ bohr for $l_{\max} = 0$ (gray), 2 (red), and 4 (blue). The well deepens with increasing $l_{\max}$ as quadrupolar and higher nuclear coupling channels open. The dotted line marks the dissociation threshold $2E_{\rm H} = -1$ Ha.

VII. DISCUSSION

A. Why $l_{\max} = 2$ is critical

The bonding orbital σ_g of H_2 concentrates electron density along the internuclear axis. In the body-frame Legendre basis, this axial concentration requires $l \geq 1$ components of $P_l(\cos\theta)$. However, gerade symmetry forces $l_1 + l_2$ even, so the lowest channel with non-zero nuclear coupling to (0,0) is at $l_{\max} = 2$. This explains three features of the convergence:

1. The $l_{\max} = 0$ result (31%) is far below He's 99.95% at $l_{\max} = 0$—isotropic averaging destroys bonding information.

2. The $l_{\max} = 1$ improvement is minimal (31% → 37%) because (1,1) has no nuclear coupling to (0,0).

3. The $l_{\max} = 2$ jump is dramatic (37% → 88%) as the quadrupolar channels (0,2) and (2,0) open.

B. The role of π channels

The σ-only expansion ($m_{\max} = 0$) uses Legendre polynomials and has no Φ-dependence. Adding π channels ($m_1 = \pm 1$, $m_2 = \mp 1$) introduces azimuthal flexibility and consistently improves D_e by ∼7 percentage points at each $l_{\max}$ (Table I). This improvement is essential: at $l_{\max} = 4$, the σ-only result (87.0%) falls below Paper 12's 92.4%, while the σ+π result (94.1%) exceeds it.

The π-channel coupling to σ channels arises exclusively through the e–e multipole expansion (the nuclear potential is diagonal in m by axial symmetry). The selection rule is $q = m_1' - m_1 = -(m_2' - m_2)$, enforced by Wigner 3-$j$ symbols in the coupling integrals.

C. The remaining 5.9% gap

The gap between 94.1% and exact has two sources:

1. **Higher partial waves.** The convergence from $l_{\max} = 3$ (87.1%) to $l_{\max} = 4$ (94.1%) with σ+π channels suggests that $l_{\max} = 5$–6 would recover further.

2. **Higher m channels.** Within the adiabatic framework, δ channels ($m_{\max} = 2$) add only ∼0.3 percentage points beyond π, indicating rapid convergence in $m_{\max}$. The 2D solver at higher $m_{\max}$ has not been tested.

D. Adiabatic vs. variational

The adiabatic approximation overestimates D_e by a remarkably constant ∼11% across all channel counts (Table I). This reflects the kinetic energy cost of angular wavefunction variation with R_e: the non-adiabatic coupling $P_{\nu\mu}(R_e) = \langle \Phi_\nu | \partial_{R_e} | \Phi_\mu \rangle$ generates off-diagonal terms in the radial kinetic energy that the adiabatic solver ignores. At moderate channel counts ($N_{\rm ch} \leq 13$), the overcounting is masked by the larger basis truncation error. At $N_{\rm ch} = 29$ ($l_{\max} = 4$, $m_{\max} = 1$), the adiabatic result exceeds $D_e^{\rm exact}$—a variational violation that signals the breakdown of the single-channel approximation. The

2D solver (Sec. VI D) resolves this by treating R_e and the angular coordinates on equal footing.

E. Fiber bundle structure

The Level 4 geometry realizes a *double-adiabatic fiber bundle*, extending the single-adiabatic bundle of Paper 13:

Outer bundle: Base = R (nuclear), Fiber = electronic states
Inner bundle: Base = R_e (hyperradial), Fiber = angular channels
Connection: $P_{\nu\mu}(R_e) = \langle \Phi_\nu | \partial_{R_e} | \Phi_\mu \rangle$

The inner connection encodes the non-adiabatic coupling between angular channels as R_e varies. Since the angular problem depends on R_e only through ρ, this connection can be computed as $P_{\nu\mu} = -(R/2R_e^2)\,\langle \Phi_\nu | \partial_\rho | \Phi_\mu \rangle$ via the Hellmann–Feynman theorem.

F. Comparison with the He hierarchy

The He hierarchy (Paper 13) has a single adiabatic layer: hyperradius as slow variable, angular coordinates as fast. The charge function $C(\alpha, \theta_{12})$ is purely angular, so the angular eigenvalue problem at fixed R_e is a standard eigenvalue problem with no parametric dependence.

In H_2, the molecular charge function depends on ρ, introducing parametric variation of the angular problem. This is structurally analogous to the Born–Oppenheimer approximation in molecular physics: nuclear coordinates R play the role of slow parameters for the electronic problem, while R_e plays the same role within the electronic problem itself.

G. Heteronuclear systems and the charge-center origin

The HeH^+ results demonstrate that the Level 4 framework generalizes naturally to heteronuclear diatomics. The key structural change is the loss of inversion symmetry, which doubles the channel count and activates odd-k multipole coupling.

The charge-center origin (Eq. 23) is not merely a computational convenience but reflects a physical asymmetry: the electronic wavefunction is concentrated around the heavier nucleus, and placing the coordinate center there reduces the angular momentum content needed to describe the lighter-nucleus contribution. The improvement—approximately one unit of $l_{\max}$—is consistent across channel counts and suggests that the charge center is the natural expansion point for asymmetric Coulomb problems in hyperspherical coordinates.

The 93.1% recovery for HeH^+ at $l_{\max} = 4$ matches the 94.1% achieved for H_2 at the same truncation, indicating that the partial-wave expansion converges at comparable rates once the origin is optimally placed. This parity supports the interpretation that the remaining ∼6% gap is dominated by higher partial waves rather than by the heteronuclear asymmetry itself.

VIII. CONCLUSION

We have introduced the Level 4 natural geometry for two-center, two-electron molecules: molecule-frame hyperspherical coordinates $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by $\rho = R/(2R_e)$.

The main results are:

1. The electron–electron cusp at $\alpha = \pi/4$, $\theta_{12} = 0$ is R-independent, resolving it as a boundary condition for all molecular geometries.

2. The angular eigenvalue problem depends on (R, R_e) only through ρ, collapsing a 2D parameter sweep to 1D.

3. The gerade–quadrupole constraint explains the dramatic jump at $l_{\max} = 2$ (31% → 80%): the first nuclear coupling from the $(0,0)$ channel requires $\Delta l = 2$.

4. π-orbital channels ($m \neq 0$) are essential, providing ∼7 percentage points of improvement at each $l_{\max}$. Without π channels, Level 4 (σ-only, 87.0%) falls below Paper 12 (92.4%); with them, it exceeds it (94.1%).

5. A variational 2D solver avoids the ∼11% systematic overestimate of the adiabatic approximation, giving rigorous bounds.

6. The framework extends to heteronuclear diatomics (HeH^+) by removing the gerade constraint, activating odd-k multipole coupling, and shifting the origin to the charge center (Eq. 23), recovering 93.1% of the exact D_e at $l_{\max} = 4$ with $\sigma+\pi$ channels.

The natural geometry hierarchy is now:

Level	System	Geometry	Best D_e
1	H	S^3 (Fock)	$< 0.1\%$ error
2	H_2^+	Prolate spheroid	0.70% error
3	He	Hyperspherical	0.05% error
4	H_2	Mol.-frame hypersp.	94.1%
4	HeH^+	Mol.-frame hypersp.	93.1%

Future work: (i) $l_{\max} \geq 5$ with $\sigma+\pi$ channels in the 2D solver; (ii) non-uniform or finite-element radial grids for improved 2D convergence; (iii) full potential energy surfaces with nuclear dynamics; (iv) heteronuclear systems beyond HeH^+ (e.g., LiH, $LiHe^+$).

[1] V. A. Fock, "Zur Theorie des Wasserstoffatoms," Z. Phys. **98**, 145 (1935).

[2] C. D. Lin, "Hyperspherical coordinate approach to atomic and other Coulombic three-body systems," Phys. Rep. **257**, 1 (1995).

[3] J. Macek, "Properties of autoionizing states of He," J. Phys. B **1**, 831 (1968).

[4] W. Kolos and L. Wolniewicz, "Improved theoretical ground-state energy of the hydrogen molecule," J. Chem. Phys. **49**, 404 (1968).

[5] H. M. James and A. S. Coolidge, "The ground state of the hydrogen molecule," J. Chem. Phys. **1**, 825 (1933).

[6] P. M. Morse and H. Feshbach, *Methods of Theoretical Physics* (McGraw-Hill, New York, 1953), Chap. 10.

[7] J. Loutey, "The Dimensionless Vacuum: Scale-Invariant Graph Topology and the Schrödinger Equation" (Paper 7 in the Geometric Vacuum series).

[8] J. Loutey, "Molecular Fock Projection: Prolate Spheroidal Lattice for Diatomic Molecules" (Paper 11 in the Geometric Vacuum series).

[9] J. Loutey, "Algebraic Two-Electron Integrals: Neumann Expansion for V_{ee} on the Prolate Spheroidal Lattice" (Paper 12 in the Geometric Vacuum series).

[10] J. Loutey, "Hyperspherical Lattice: Two-Electron Atoms as Coupled-Channel Graphs" (Paper 13 in the Geometric Vacuum series).

[11] D. M. Bishop and L. M. Cheung, "A theoretical investigation of HeH^+," J. Mol. Spectrosc. **75**, 462 (1979).